

CURRICULUM VITAE (Detailed)

****Research Statement****

I am Prof. Dr. R. Dinakaran Michael, formerly, Dean of Life Sciences at Vels University, Chennai, India and Head of PG Zoology, The American College, Madurai. My research interests (**Annexure 1**) are in the field of fish immunology, immunostimulation and immunotoxicology. I have more than 25 years of experience in directing the Research Centre for Fish Immunology at various institutions in India. I have also organized and participated in several international and national conferences, workshops and training programs on immunological techniques and aquaculture.

My research aims to understand the immune system of fish and its modulation by various natural and synthetic compounds, as well as environmental factors such as stress, pollution and pathogens. I have used different fish models such as *Oreochromis mossambicus*, *Cyprinus carpio*, *Channa striata* and *Labeo rohita* to study the effects of immunostimulants such as plant extracts, azadirachtin and tannery effluent on their non-specific immune responses and disease resistance. I have also investigated the immunotoxic effects of heavy metals such as chromium and cadmium on fish immune cells and organs.

My research has contributed to the advancement of knowledge in fish immunology and has implications for aquaculture industry, environmental management and perhaps human health. I have published more than 50 peer-reviewed papers in reputed journals such as Fish & Shellfish Immunology, Indian Journal of Experimental Biology, Aquaculture Research and Ecotoxicology and Environmental Safety. I have also authored two books and several book chapters on fish immunology. I have received many awards and honours for my research achievements, such as the UGC Emeritus Fellow, FAO Expert/Consultant for Fish Immunology, Scientist Exchange Fellowship Award of the Indian National Science Academy- DFG, Germany, State Best Teacher Award of Tamil Nadu Government and John Humphrey Fellowship of British Society for Immunology.

I worked on several research projects funded by various agencies such as UGC, DST, DBT, OASTC (MoES)

My recent projects were all on Immunostimulation by marine algae in fish. Earlier I was looking at the Immunotoxicity of heavy metals in fish.

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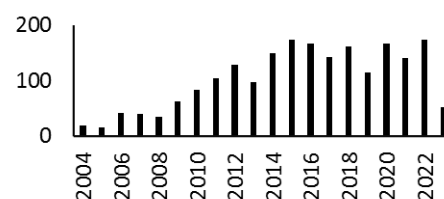


<https://scholar.google.com/citations?user=4YPwzIMAAAJ&hl=en>



<https://www.scopus.com/authid/detail.uri?authorId=7102954651>

Number of citations



Cited by

	All	Since 2018
Citations	2113	812
h-index	21	15
i10-index	33	23

I am also actively involved in teaching and mentoring undergraduate and postgraduate students, as well as research scholars in the field of Immunology. I have supervised more than 12 PhD students and 35 MPhil students who have successfully completed their degrees. I have also developed several innovative courses and curricula on Immunology and Microbiology at various levels.

****Experience****

Teaching (Annexure 2)

More than 4 decades of teaching Immunology and Immunotechnology and Biochemistry etc. at MSc and MPhil levels in the American College and Lady Doak College, Madurai, India; Vels University, Chennai; Bellevue University, Bellevue, NE, USA and Oberlin College, Oberlin, OH, USA

Research

- Papers published in indexed national and international Journals – 53 (Annexure 3)
- Patents filed – 4 (Annexure 4)
- Abstracts published in Proceedings of National/ International Conferences–100 (International-27) (Annexure 5)
- Invited Lectures in India and abroad–45 (Abroad-10) (Annexure 6)
- Major research projects carried out –8 (Annexure 7)
- PhD students guided -12 (Annexure 8)
- MPhil students guided-32 (Annexure 9)
- Total Research grants received – About 2 Crores from Volkswagen Foundation (Germany), United Board (USA), DBT, UGC, DST, OASTC (MoES) etc.
- Membership in Professional Societies-7 (Annexure 10)

****Academic qualifications****

Degree	Institution	Year	Distinction
Ph.D. (Immunology)	Madurai Kamaraj University	1986	'Commended'
P.G. Diploma in Hr. Edn.	Madurai Kamaraj University	1979	1 st rank
M.Phil. (Animal Physiology)	Madurai Kamaraj University	1977	Varsity 1 st rank
Certificate in Biochemistry	Madurai Kamaraj University	1974	Distinction
M.Sc. (Zoology)	University of Madras	1966	Varsity 3 rd rank

****Administration and Organization****

15 years of heading two P.G. Departments (1. Zoology and 2. Immunology & Microbiology) at the American College, Madurai, India. Introduced new and innovative UG and PG programmes on Immunology and Microbiology. More than 30 years of directing the Research Centre for Fish Immunology at the American College, Madurai, later at Lady Doak College, Madurai and now recently at Vels University, Chennai. Organized one international conference, four National training workshops on Immunological techniques and 3 State level workshops, two on Immunology and the third on Biostatistics.

****Community service****

Basic Health Care and Immunization

Coordinated a tribal outreach program involving the PG students of our department for about 12 years sponsored by United Board, New York, USA. Under this Service learning program for sensitizing the young people to the needs of the marginalized sections of our society, about 1200 tribal infants and children were vaccinated and boosted (therefore, protected) against Diphtheria, Pertussis, Tetanus (Triple antigen) and Polio and hundreds of tribal adults were screened for diseases by periodic visits by students to tribal pockets in Thandikudi hills.

****International assignments****

2002 Spring Visiting Scientist, Hannover Veterinary University, Hannover, Germany
2001 Spring Visiting Scientist, Hannover Veterinary University, Hannover, Germany
1999 Fall Visiting Professor, Oberlin College, Oberlin, Ohio, USA
1999 Spring Visiting Scientist, Hannover Veterinary University, Hannover, Germany
1998 Fall Visiting Scholar, Oberlin College, Oberlin, Ohio, USA
1990-1991 Visiting Professor, Bellevue University, Bellevue, Nebraska, USA

****Awards/honours/distinctions****

2011 Selected as UGC Emeritus Fellow by University Grants Commission, New Delhi
2003 Selected as FAO Expert/Consultant for Fish Immunology by Food and Agriculture Organization (FAO) of the United Nations, Rome, Italy.
1999 Visiting Professorship of Oberlin-Shansi Memorial association to offer a course on Aquaculture in Oberlin College, Oberlin, Ohio, USA
1999 Recipient of Scientist Exchange Fellowship Award of the Indian National Science Academy- DFG, Germany to visit Hannover Veterinary University, Germany.
1998 Invited by American Fisheries Society, Fish Health Section, to present a paper in the III International Conference on Aquatic Animal Health held at Baltimore, USA
1998 Visiting Scholar Fellowship of Oberlin-Shansi Memorial Association, Oberlin College, Oberlin, Ohio.
1997 Recipient of State Best Teacher Award (College-University level - Dr. Radhakrishnan medal) of Tamil Nadu Government.
1996 Recipient of Fr. T.A. Mathias award of All India Association for Christian higher Education, New Delhi, given to distinguished and innovative teachers.
1996 Invited by Australian Fisheries Society to present papers in Second World Fisheries Congress held at Brisbane, Australia.
1995 Invited by China Fisheries Society to present papers in IV Asian Fisheries Forum at Beijing, China. Support also provided by Department of Science and Technology, Govt. of India, New Delhi.
1995 Invited by Hongkong Baptist University to give a lecture on Fish Immunology at the University, Hong Kong.
1994 Invited by International Society of Developmental & Comparative Immunology (ISDCI) presented a paper in VI ISDCI Congress at the Netherlands.
1990 Asia-North America Faculty Exchange fellowship (one year of visiting Professorship in Bellevue University, Bellevue, Nebraska, USA to teach Immunology etc.) of United Board for Christian Higher education in Asia, New York, USA.

1989 "John Humphrey Fellowship" of British Society for Immunology, London for developing an Immunology Research Laboratory in India.

****Countries visited****

USA, Canada, The Netherlands, UK, Germany, Australia, China, Hong Kong, Malaysia, Singapore, and Taiwan on teaching/research/lecture assignments.

Annexure 1

Research Interest

The main areas of our research interest include developing environment-friendly prophylactic and therapeutic immunostimulants from terrestrial and marine plants for culture fish. Another area is heavy metal-induced immunosuppression in fish and this aspect has relevance to extensive cultures in large water bodies which are more often receive polluted waters from rivers and canals which are polluted by industrial effluents. Stress-induced immunomodulation due to social stressors (e.g., crowding, monosex cultures) is another priority area of research in this centre.

Plant-derived immunostimulants

Aquaculture faces substantial annual loss of production due to diseases. Antibiotics and chemo-therapeutic agents are being used to control microbial diseases in aquaculture systems. These agents are not only expensive, but they are also environmental pollutants. Further, they often result in the development of antibiotic resistant strains of microbes. In this context, terrestrial and marine plant-derived, immunostimulating compounds offer promising results due to its broad spectral activity and cost-effectiveness. India has been bestowed with enormous wealth of traditional terrestrial medicinal plants and marine plants. Hence the centre aims at studying the immunostimulatory and antibacterial properties of these plants' extracts in fish for developing the processes and the products for wider application of them as immunoprophylactic and therapeutics in aquaculture in this country.

Heavy metal induced immunomodulation

Aquatic environment of fish is in close contact with numerous pollutants. Aquatic pollutants such as heavy metals modulate (most of them suppress) the immune system of fish, thus increasing the host susceptibility to infectious pathogens. So, one of the aims of the centre is to investigate the effect of heavy metals such as chromium, cadmium, mercury and other common pollutants on the immune status of fish. This assessment of the immune status of fish is not only helpful as the source of basic immunotoxicology information but also as immuno-indicators for monitoring fish health in extensive fish cultures in large lakes and ponds.

Stress-induced immunomodulation

Intensive aquaculture practices where culture fish is usually overcrowded often result in stress-mediated immunosuppression in fish, leading to sudden outbreak of fish diseases. This laboratory aims at finding the mechanism behind stress-mediated immunosuppression in fish. Another related area of interest is the effect of complex social interactions among the culture fish communities (all male, all female or mixed population) on the immune system of fish. This approach can help fish farmers to devise species-friendly management practices thereby preventing heavy loss due to outbreak of diseases.

Annexure 2
Teaching

No	Designation & Office address	Period		Nature of duties/responsibilities
		From	To	
I.	Dean of Life Sciences Vels University, Pallavaram Chennai-117	July 2012	Sep 2019	Coordinating and Directing four Departments in Life Sciences, Research in Fish Immunology
II.	Professor of Immunology and Director, School of Life Sciences Vels University, Pallavaram Chennai- 117	July 2010	June 2012	Coordinating and Directing four Departments in Life Sciences, Research in Fish Immunology and Teaching Immunology for PG students
III.	Professor of Immunology & Director, Centre for Fish Immunology, Lady Doak College, Madurai-2	June 2003	June 2010	Research in Fish Immunology and Teaching Immunology for PG and MPhil students
IV.	Professor and Head, PG Departments of Zoology & Immunology and Microbiology The American College, Madurai-2	1985	2003	Research and Teaching Immunology and Fish Immunology
V.	Visiting Professor, Oberlin College, Oberlin, Ohio, USA	1999 Fall Sem		Teaching a course on Aquaculture
VI.	Visiting Scientist (INSA-DFG), Dept. of Fish Pathology, Hannover Veterinary University, Hannover, Germany	1999 Mar	1999 June	Collaborative research
VII.	Visiting Scientist, Oberlin College, Oberlin, Ohio, USA	1998 Fall Sem		Designing a major course on Aquaculture
VIII.	Visiting Professor Bellevue University, Bellevue, Nebraska, USA.	1990 Aug	1991 May	Teaching Biology, Nutrition, Immunology
IX.	Professor of Zoology, American College, Madurai-2	1980	1985	Research and Teaching Immunology and Biochemistry
X.	Assistant Professor of Zoology, American College, Madurai-2	1968	1980	Research and Teaching Zoology, Biochemistry, Immunology etc
XI.	Demonstrator in Zoology, Bishop Heber College, Trichy.	1967	1968	Teaching Biology
XII.	Survey Assistant in CMFRI, Ramnad Dt.	1966	1967	Training in Fishery Survey and Statistics

Annexure 3 Publications

Books

1. 2018 Immunological Techniques Using Fish Model- A Laboratory Manual (Student Version)- Notion Press, Chennai, India ISBN:978-1-64324-645-1
2. 2018 Immunological Techniques Using Fish Model- A Laboratory Manual (Institutional Version): Notion Press, Chennai, India ISBN: 978-1-64429-633-2

Journals and chapters

1.	2022	Subramani P.A., Shaik F.B., Michael R.D. , Panati K., Narala V.R., Thiamine: A Natural Peroxisome Proliferator-Activated Receptor Gamma (PPAR-γ) Activator. Letters in Drug Design and Discovery: 19(10) 888-896. Doi: 10.2174/1570180819666220127121403
2.	2019	Shalini K.S., Yengkhom O., Subramani P.A., Michael R.D. , Polysaccharide fraction from the Indian mistletoe, <i>Dendrophthoe falcata</i> (L.f.) Ettingsh enhances innate immunity and disease resistance in <i>Oreochromis niloticus</i> (Linn.). Fish and Shellfish Immunology: 88 407-414. Doi: 10.1016/j.fsi.2019.03.008
3.	2019	Shalini K.S., Yengkhom O., Subramani P.A., Michael R.D. , Polysaccharide fraction of the hemi-parasitic mistletoe, <i>Dendrophthoe falcata</i> (L) ettingsh leaves enhances innate immune responses and disease resistance in <i>Oreochromis niloticus</i> (Linnaeus). International Journal of Research in Pharmaceutical Sciences: 10(1) 56-64. Doi: 10.26452/ijrps.v10i1.1780
4.	2019	Yengkhom O., Shalini K.S., Subramani P.A., Michael R.D. , Stimulation of non-specific immunity, gene expression, and disease resistance in Nile Tilapia, <i>Oreochromis niloticus</i> (Linnaeus, 1758), by the methanolic extract of the marine macroalga, <i>Caulerpa scalpelliformis</i> . Veterinary World: 12(2) 271-276. Doi: 10.14202/vetworld.2019.271-276
5.	2018	Yengkhom O., Shalini K.S., Subramani P.A., Michael R.D. , Non-specific immunity and disease resistance are enhanced by the polysaccharide fraction of a marine chlorophycean macroalga in <i>Oreochromis niloticus</i> (Linnaeus, 1758). Journal of Applied Ichthyology: 34(3) 556-567. Doi: 10.1111/jai.13606
6.	2017	Michael RD. Aquaculture in India, the Second Largest Producer of Farmed Fish. Int J Oceanogr Aquac 2017, 1(3): 000114.
7.	2017	Priyadarshini S.K., Subramani P.A., Michael R.D. , Modulation of the innate immune responses in the striped snakehead murrel, <i>Channa striata</i> upon experimental infection with live and heat killed <i>Aeromonas hydrophila</i> . Open Veterinary Journal: 7(2) 157-164. Doi: 10.4314/ovj.v7i2.13
8.	2017	Subramani P.A., Michael R.D. , Prophylactic and prevention methods against diseases in aquaculture. Fish Diseases: Prevention and Control Strategies: 81-117. Doi: 10.1016/B978-0-12-804564-0.00004-1
9.	2016	Subramani P.A., Daniels Gnanamuthu A.A., Michael R.D. , Neutrophil activity affects <i>Oreochromis mossambicus</i> (Peters, 1852) antibody production against heat-killed <i>Aeromonas hydrophila</i> vaccine. Journal of Applied Ichthyology: 32(6) 1113-1117. Doi: 10.1111/jai.13123
10.	2016	Subramani P.A., Narasimha R.V., Balasubramanian R., Narala V.R., Ganesh M.R., Michael R.D. , Cytotoxic effects of <i>Aeromonas hydrophila</i> culture supernatant on peripheral blood leukocytes of Nile tilapia (<i>Oreochromis niloticus</i>): Possible presence of a secreted cytotoxic lectin. Fish and Shellfish Immunology: 58 604-611. Doi: 10.1016/j.fsi.2016.09.061
11.	2016	Rajendran P., Subramani P.A., Michael R.D. , Polysaccharides from marine macroalga, <i>Padina gymnospora</i> improve the nonspecific and specific immune responses of <i>Cyprinus carpio</i> and protect it from different pathogens. Fish and Shellfish Immunology: 58 220-228. Doi: 10.1016/j.fsi.2016.09.016
12.	2016	Mahendran R., Jeyabaskar S., Sitharaman G., Michael R.D. , Paul A.V., Computer-aided vaccine designing approach against fish pathogens <i>Edwardsiella tarda</i> and <i>Flavobacterium columnare</i> using bioinformatics softwares. Drug Design, Development and Therapy: 10 1703-1714. Doi: 10.2147/DDDT.S95691
13.	2016	Kirubakaran C.J.W., Subramani P.A., Michael R.D. , Methanol extract of <i>Nyctanthes arbortristis</i> seeds enhances non-specific immune responses and protects <i>Oreochromis mossambicus</i> (Peters) against <i>Aeromonas hydrophila</i> infection. Research in Veterinary Science: 105 243-248. Doi: 10.1016/j.rvsc.2016.02.013
14.	2016	Subramani P.A., Mahendran R., Michael R.D. , Molecular docking and dynamics simulation of <i>Vibrio anguillarum</i> aspartate semialdehyde dehydrogenase with natural product caulerpin. Letters in Drug Design and Discovery: 13(3) 255-261. Doi: 10.2174/1570180812666150730221627
15.	2016	Balasubramanian R., Michael R.D. , Immunostimulatory effects of N-Oxide-Quaternary alkaloid fraction of a marine Chlorophycean macroalga in the striped murrel, <i>Channa striata</i> (Bloch). Aquaculture Research: 47(2) 591-604. Doi: 10.1111/are.12518

16.	2016	Muthulakshmi M., Subramani P.A., Michael R.D. , Immunostimulatory effect of the aqueous leaf extract of <i>Phyllanthus niruri</i> on the specific and nonspecific immune responses of <i>Oreochromis mossambicus</i> Peters. Iranian Journal of Veterinary Research: 17(3) 200-202. Doi:
17.	2015	Subramani P.A., Hameed B., Michael R.D. , Effect of UV-B radiation on the antibody response of fish - Implication on high altitude fish culture. Journal of Photochemistry and Photobiology B: Biology: 143 1-4. Doi: 10.1016/j.jphotobiol.2014.12.021
18.	2015	Subramani, P. A.; Narala, V. R.; Michael, R. D. ; Lomada, D.; Reddy, M. C., Molecular docking and simulation of Curcumin with Geranylgeranyl Transferase1 (GGTase1) and Farnesyl Transferase (FTase). Bioinformation. 11, 248-253.
19.	2014	Devasree L.D., Binuramesh C., Michael R.D. , Immunostimulatory effect of water soluble fraction of <i>Nyctanthes arborescens</i> leaves on the immune response in <i>Oreochromis mossambicus</i> (Peters). Aquaculture Research: 45(10) 1581-1590. Doi: 10.1111/are.12104
20.	2013	Richard B.R., Omita Y., Michael R.D. , Immunostimulation by Vitamin A in Mozambique Tilapia, <i>Oreochromis mossambicus</i> (Peters). International Journal of Frontiers in Science and Technology 1(4):1-12
21.	2012	L.D. Devasree, C. Binuramesh, Michael R.D. , Immunostimulatory effect of water soluble fraction of <i>Nyctanthes arborescens</i> leaves on the immune response in <i>Oreochromis mossambicus</i> . Aquaculture Research online doi:10.1111/are/12104
22.	2012	Michael R.D. , Priyadarshini S.K., A reliable method for repetitive bleeding in striped murrel, <i>Channa striata</i> (Bloch). Aquaculture Research: 43(11) 1738-1740. Doi: 10.1111/j.1365-2109.2011.02987.x
23.	2011	Binuramesh C., Michael R.D. , Diel variations in the selected serum immune parameters in <i>Oreochromis mossambicus</i> . Fish and Shellfish Immunology: 30(3) 824-829. Doi: 10.1016/j.fsi.2011.01.004
24.	2010	Kirubakaran C.J.W., Alexander C.P., Michael R.D. , Enhancement of non-specific immune responses and disease resistance on oral administration of <i>Nyctanthes arborescens</i> seed extract in <i>Oreochromis mossambicus</i> (Peters). Aquaculture Research: 41(11) 1630-1639. Doi: 10.1111/j.1365-2109.2010.02516.x
25.	2010	Alexander C.P., John Wesley Kirubakaran C., Michael R.D. , Water soluble fraction of <i>Tinospora cordifolia</i> leaves enhanced the non-specific immune mechanisms and disease resistance in <i>Oreochromis mossambicus</i> . Fish and Shellfish Immunology: 29(5) 765-772. Doi: 10.1016/j.fsi.2010.07.003
26.	2007	Prabakaran M., Binuramesh C., Steinhagen D., Dinakaran Michael R.D. , Immune response in the tilapia, <i>Oreochromis mossambicus</i> on exposure to tannery effluent. Ecotoxicology and Environmental Safety: 68(3) 372-378. Doi: 10.1016/j.ecoenv.2006.11.016
27.	2007	Christyapita D., Divyagnaneswari M., Dinakaran Michael R.D. , Oral administration of <i>Eclipta alba</i> leaf aqueous extract enhances the non-specific immune responses and disease resistance of <i>Oreochromis mossambicus</i> . Fish and Shellfish Immunology: 23(4) 840-852. Doi: 10.1016/j.fsi.2007.03.010
28.	2007	Divyagnaneswari M., Christyapita D., Michael R.D. , Enhancement of nonspecific immunity and disease resistance in <i>Oreochromis mossambicus</i> by <i>Solanum trilobatum</i> leaf fractions. Fish and Shellfish Immunology: 23(2) 249-259. Doi: 10.1016/j.fsi.2006.09.015
29.	2006	Samuel Sudhakaran D., Sirekha P., Devasree L.D., Prem Singh S., Michael R.D. , Immunostimulatory effect of <i>Tinospora cordifolia</i> Miers leaf extract in <i>Oreochromis mossambicus</i> . Indian Journal of Experimental Biology: 44(9) 726-732. Doi:
30.	2006	Binuramesh C., Prabakaran M., Steinhagen D., Michael R.D. , Effect of sex ratio on the immune system of <i>Oreochromis mossambicus</i> (Peters). Brain, Behavior, and Immunity: 20(3) 300-308. Doi: 10.1016/j.bbi.2005.09.002
31.	2006	Prabakaran M., Binuramesh C., Steinhagen D., Michael R.D. , Immune response and disease resistance of <i>Oreochromis mossambicus</i> to <i>Aeromonas hydrophila</i> after exposure to hexavalent chromium. Diseases of Aquatic Organisms: 68(3) 189-196. Doi: 10.3354/dao068189
32.	2005	Binuramesh C., Prabakaran M., Steinhagen D., Michael R.D. , Effect of chronic confinement stress on the immune responses in different sex ratio groups of <i>Oreochromis mossambicus</i> (Peters). Aquaculture: 250(44958) 47-59. Doi: 10.1016/j.aquaculture.2005.03.043
33.	2004	Steinhagen D., Helmus T., Maurer S., Michael R.D. , Leibold W., Scharsack J.P., Skouras A., Schuberth H.-J., Effect of hexavalent carcinogenic chromium on carp <i>Cyprinus carpio</i> immune cells. Diseases of Aquatic Organisms: 62(44958) 155-161. Doi: 10.3354/dao062155
34.	2003	Michael R.D. , Scope for using medicinal plant extracts as immunostimulants in finfish aquaculture. 'Aquaculture Medicine' published by Cochin University of Science and technology. ISBN: 81-900724-1-2
35.	2003	Scharsack J.P., Steinhagen D., Kleczka C., Schmidt J.O., Körting W., Michael R.D. , Leibold W., Schuberth H.J., Head kidney neutrophils of carp (<i>Cyprinus carpio</i> L.) are functionally modulated by the haemoflagellate <i>Trypanoplasma borreli</i> . Fish and Shellfish Immunology: 14(5) 389-403. Doi: 10.1006/fsim.2002.0447
36.	2003	Scharsack J.P., Steinhagen D., Kleczka C., Schmidt J.O., Körting W., Michael R.D. , Leibold W., Schuberth H.J., The haemoflagellate <i>Trypanoplasma borreli</i> induces the production of nitric oxide, which is associated

		with modulation of carp (<i>Cyprinus carpio</i> L.) leucocyte functions. Fish and Shellfish Immunology: 14(3) 207-222. Doi: 10.1006/fsim.2002.0430
37.	2002	Chandran M.R., Aruna B.V., Logambal S.M., Michael R.D. , Immunisation of Indian major carps against <i>Aeromonas hydrophila</i> by intraperitoneal injection. Fish and Shellfish Immunology: 13(1) 1-9. Doi: 10.1006/fsim.2001.0374
38.	2001	Aruna B.V., Chandran M.R., Dinakaran Michael R.D. , Logambal S.M., Surface characters and extracellular toxins involved in the pathogenesis of <i>Aeromonas hydrophila</i> . Indian Journal of Experimental Biology: 39(10) 1048-1054. Doi:
39.	2000	Logambal S.M., Michael R.D. , Immunostimulatory effect of azadirachtin in <i>Oreochromis mossambicus</i> (Peters). Indian Journal of Experimental Biology: 38(11) 1092-1096. Doi:
40.	2000	Arunkumar R.I., Rajasekaran P., Michael R.D. , Differential effect of chromium compounds on the immune response of the African mouth breeder <i>Oreochromis mossambicus</i> (Peters). Fish and Shellfish Immunology: 10(8) 667-676. Doi: 10.1006/fsim.2000.0281
41.	2000	Logambal S.M., Venkatalakshmi S., Michael R.D. , Immunostimulatory effect of leaf extract of <i>Ocimum sanctum</i> Linn. in <i>Oreochromis mossambicus</i> (Peters). Hydrobiologia: 430(44986) 113-120. Doi: 10.1023/a:1004029332114
42.	2000	Rajavarthini P.B., Arunkumar R.I., Michael R.D. , Partial characterization of serum immunoglobulins of <i>Oreochromis mossambicus</i> . Indian Journal of Experimental Biology: 38(6) 549-553. Doi:
43.	1998	Michael R.D. , Quinn C.G., Venkatalakshmi S., Modulation of humoral immune response by ascorbic acid in <i>Oreochromis mossambicus</i> (Peters). Indian Journal of Experimental Biology: 36(10) 1038-1040. Doi:
44.	1998	Ravikumar M., Michael R.D. , Influence of group density and sex ratio on the immune response in the tilapia <i>Oreochromis mossambicus</i> (Peters). Current Science: 74(6) 534-537. Doi:
45.	1996	Rajavarthini P.B., Michael R.D. , The effect of Monocrotophos (Nuvacron) on the antibody response to bovine serum albumin in <i>Oreochromis mossambicus</i> (Peters). J. Aqua. Trop. 11: 27-32
46.	1995	Sudhan T., Michael R.D. , Modulation of humoral immune response by tannery effluent in <i>Oreochromis mossambicus</i> (Peters). Indian journal of experimental biology: 33(10) 793-795. Doi:
47.	1994	Michael R.D. , Rajasekaran P., I.P5 Differential effect of chromium compounds on the immune response in <i>Oreochromis mossambicus</i> (peters). Developmental and Comparative Immunology: 18(SUPPL. 1) -. Doi: 10.1016/0145-305X(94)90220-8
48.	1994	Michael R.D. , Priscilla A.S., Diel variations in antibody response to bovine serum albumin in <i>Oreochromis mossambicus</i> (Peters). Indian Journal of Experimental Biology: 32(7) 474-477. Doi:
49.	1994	Michael R.D. , S.D. Srinivas, K. Sailendri, Muthukkaruppan V.R., A rapid method for repetitive bleeding in fish. Ind. J. Exp. Biol. 32: 838- 839
50.	1984	Michael R.D. , Muthukkaruppan V.R., A simple method for serial bleeding of lizards. Developmental and Comparative Immunology: 8(1) 209-212. Doi: 10.1016/0145-305X(84)90025-9
51.	1977	Dwarakanath S.K., Michael R.D. , Humidity reactions in the pentatomid insect <i>Chrysocoris purpureus</i> (westwood). Monitore Zoologico Italiano - Italian Journal of Zoology: 11(44958) 1-8. Doi: 10.1080/00269786.1977.10736285

Annexure 4

Patents

1. A potent immunostimulant chemically separated from a marine macroalga (seaweed) for prophylactic (preventive) health management in finfish aquaculture. Inventors: **Dr. R. Dinakaran Michael** and Ms. Ramalakshmi Balasubramanian. Patent registration No: 2271/CHE/2012
2. An efficacious marine plant - derived immunostimulant for preventing diseases in carps and other fishes in aquaculture. Inventors: **Dr. R. Dinakaran Michael** and Ms. Priyatharsini Rajendran. Patent registration No: 524/CHE/2015
3. Seaweed-derived immunostimulant for protecting murrels (Snake head fish) and other culture fishes from diseases. Inventors: **Dr. R. Dinakaran Michael** and Ms. S. Kalaivani Priyadarshini. Patent registration No: 4022/CHE/2015
4. A Polysaccharide Immunostimulant for preventing diseases in finfish aquaculture. Inventors: **Dr. R. Dinakaran Michael** and Ms. Priyatharsini Rajendran. Patent Application number No: 201641027311 dated 10-8-2016
5. Mistletoe-derived polysaccharide immunostimulant for preventing infectious diseases in finfish aquaculture. Inventors: **Dr. R. Dinakaran Michael**, K. S. Shalini, A. S. Parasuraman and Omita Yengkhom (under Processing)
6. The Idea of Using Fish Model to Demonstrate Immunological Principles in Immunology Laboratory Courses of Life Science Programmes in Higher Educational Institutions. Inventor: **Dr. R. Dinakaran Michael** (under Processing)

Annexure 5

Papers presented in International and National Conferences

****International Conferences Abroad****

2008 The 7th International Symposium on Diseases in Asian Aquaculture at Taipei

1. Immunoprophylactics of plant origin for fin fish Aquaculture.
2. Immunostimulatory effect of chloroform extract of *Nyctanthes arbor-tristis* seeds in *Oreochromis mossambicus*.

2005 VI Symposium on Diseases in Asian Aqua culture, Colombo, Srilanka.

3. Effect of *Eclipta alba* leaf extracts on the immune response of *Oreochromis mossambicus*.
4. Immunomodulatory activity of *Solanum trilobatum* leaf extracts in *Oreochromis mossambicus*.
5. Immunostimulatory effect of water-soluble fraction of *Tinospora cordifolia* leaves on *Oreochromis mossambicus*.
6. Immunomodulation by few Indian medicinal plants in *Oreochromis mossambicus*.

2004 The 7th Asian Fisheries Forum 04, The Triennial meeting of the Asian Fisheries Society at Penang, Malaysia.

7. Effect of *Solanum trilobatum* leaf extracts on the bacterial pathogens and immune response of *Oreochromis mossambicus*.
8. Immunostimulatory effect of medicinal plant *Nyctanthes arbor-tristis* in *Oreochromis mossambicus*.
9. Immunostimulatory effect of Selenium in Tilapia, *Oreochromis mossambicus*.

1998 3rd International Symposium on Aquatic Animal Health Organized by American Fisheries Society (Fish Health Section) at Baltimore.

10. Immunostimulatory effect of leaf extract of *Ocimum sanctum* Linn. in *Oreochromis mossambicus* (Peters)

1994 2nd World Fisheries Congress, Brisbane, Queensland, Australia.

11. Immunoindicators of environmental stress/pollution and of disease-outbreak in aquaculture.
12. Differential immunomodulatory effect of nickel compounds in *Oreochromis mossambicus* (Peters).

1995 4th Asian Fisheries Forum, Beijing, China

13. Effect of UV-b radiation on the antibody response in *Oreochromis mossambicus* (Peters)
14. Modulatory effect of the organic and inorganic forms of Mercury on the humoral immune response of *Oreochromis mossambicus* (Peters) to a soluble protein antigen

15. Relative efficacy of immersion and hyperosmotic infiltration methods of immunization in fish.

1994 6th Congress International Society of Developmental and Comparative Immunology, Wageningen, The Netherlands

16. Differential effect of Chromium compounds on the immune response in *Oreochromis mossambicus* (Peters).

1983 International Congress of Immunology, Kyoto, Japan.

17. Effect of splenectomy in the lizard *Calotes versicolor* differential response to differential forms of antigen.

****International Conferences in India****

2011 Eighth Symposium on Diseases in Asian Aquaculture, Mangalore on 21-25 November, 2011

18. Marine macroalgal immunostimulants- An alternative to finfish vaccines?
19. Inducible non-specific immune responses in striped murrel, *Channa striata*.
20. Effect of the macroalga, CFIMA 04 on the expression of immune response genes in the head kidney cells of the carp, *Cyprinus carpio*.

2006 Fourth International Symposium on Biocontrol and Biotechnology, Lady Doak College, Madurai

21. Immunostimulation by water soluble fraction of *Nyctanthes arbor-tristis* L. leaves in *Oreochromis mossambicus* (Peters).
22. Oral administration of *Solanum trilobatum* enhances specific immunity and disease resistance in *Oreochromis mossambicus*.
23. Enhancement of Nonspecific immunity and Disease resistance in *Oreochromis mossambicus* by *Eclipta alba*.
24. *Tinospora cordifolia* – A potential oral immunostimulant for *Oreochromis mossambicus*.

2003 International Conference on Disease Management for Sustainable Fisheries at Trivandrum

25. In search of immunostimulants of plant origin for finfish aquaculture.
26. Immunomodulatory effect of Chromium VI on the immune response of *Oreochromis mossambicus*.
27. Effect of confinement and sex ratio on the immune response in *Oreochromis mossambicus*.

****National Conferences****

2018 National conference on “Bioactive marine natural products and their therapeutic applications” Sathyabama Institute of Science and Technology, Chennai-119, on 24-25th January 2018

28. Marine Macroalgae- An Ideal Source of Immunostimulants for Finfish Health Management

2017 National conference on “Recent Advances in Zebra Fish Immunology research: An ideal model For Human Health”, Sathyabama Institute of Science and Technology, Chennai-119, on 20-21 Sep 2017

29. Fish- an ideal model for Immunology Teaching and Research

2011 9th Indian Fisheries Forum, Chennai on 19-23 December, 2011

30. Marine Macroalgae – An Excellent Source of Immunostimulants for Finfish.

2011 National Workshop on Research advances in Fish Vaccines and Prophylactics, Tamilnadu Veterinary and Animal Sciences University, Fisheries College and Research Institute, Thoothukudi on 14-15 February, 2011

31. Plant (Marine and Terrestrial- Derived Immunostimulants – A Viable Alternative to Fish Vaccines
32. Disease protective and antibacterial properties of a Phaeophycean macroalga
33. Immunomodulatory properties of a Phaeophycean macroalga in the carp, *Cyprinus carpio*
34. Enhancement of non-specific immune responses and disease protective properties of Phaeophycean Macroalga (CFI-MA 01) in *Oreochromis mossambicus* (PETERS)
35. Inducible non-specific immune responses in *Channa striata* (Bloch)
36. Diel variation of humoral nonspecific immune mechanisms in *Channa striatus* (Bloch)
37. Immunostimulatory and antibacterial activities of a chlorophycean macroalga in *Channa striatus* (Bloch)

2010 Second National Conference on Bioprospecting of Marine Resources with Special Reference to Marine Natural Products and Drug Discovery, School of Marine Sciences, Department of Oceanography and Coastal Area Studies, Alagappa University, Thondi Campus, Ramanathapuram on 25-27 August, 2010

38. A study on the disease protective properties of a Phaeophycean macroalga CFI-MA 04 in *Cyprinus carpio*.
39. Immune enhancement by methanol extract of a marine macroalga (CFI-MA 01) in spotted murrel, *Channa punctatus*.
40. Bioactive Potential of marine macroalgae in terms of antibacterial and disease protective activities in fish

2010 National Conference on Environment, Climate change and Aquaculture, Andhra University, Vishakapatnam, Aug 2010

41. Plant - Derived Immunostimulants – A Viable Alternative To Fish Vaccines

2010 41st Annual Conference on Organismal and Functional Genomics.

**School of Biological Sciences, Madurai Kamaraj University, Madurai on
6th March, 2010**

42. Siderophores and Lipopolysaccharides of *Aeromonas hydrophila* as vaccine for *Oreochromis mossambicus*.
43. Rhythmicity in nonspecific immune mechanisms of *Channa striatus* (Bloch).
44. Evaluation of non-specific immune mechanisms in *Channa striatus* (Bloch) on experimental infection with live and killed *Aeromonas hydrophila*.

2006 National Seminar on Biomedicine in Aquaculture Centre for Marine Science and Technology, MS University, Kanyakumari district, Tamil Nadu on 17-18 March.

45. Immunostimulants of plant origin for finfish aquaculture.

2006 National conference on Aquatic Resources, Aquaculture and Aquashow, St. Xavier's College, Palayamkottai, Tamil Nadu, 15-17 February.

46. Aquatic Chicken for Indian Aquaculture?

2006 National Conference on Frontiers in Biotechnology and molecular medicine. Noorul Islam college of arts and science. Kumaracoil, Thuckalay, K.K. Dist, Tamil Nadu, on 17-18 February.

47. Antibody Engineering

2005 Indian Fisheries Forum Karnataka Veterinary, Animal & Fisheries Sciences University, Hebbal, Bangalore-24, on 8-12 November.

48. Plant-Derived Immunostimulants for Finfish Aquaculture.

2004 National seminar on Recent advances in Biology. Jamal Mohamed College, Trichy, on 24th August.

49. Hybridoma Technology

2004 National seminar on Recent advances in Biochemistry Ayyanadar Janaki Ammal College, Sivakasi on 20th August

50. Genetic basis of Immunoglobulin Structure and Diversity

2004 National seminar on Recent advances in Biology. Jamal Mohamed College Trichy on 5th June.

51. Genetically Engineered Vaccines

2001 National workshop on Aquaculture Medicine, Centre for Fish Disease Diagnosis and Management, Cochin University of Science and Technology, Cochin

52. Immunostimulation by leaf extract of *Crossandra infundibuliformis* in *Oreochromis mossambicus* (Peters).
53. Differential immunomodulatory effect of chemical and social stressors in *Oreochromis mossambicus* (Peters).
54. Immunostimulatory effect of *Ocimum sanctum* leaf extract administered by immersion in *Oreochromis mossambicus*

55. Scope for using medicinal plant extracts as immunostimulants in finfish aquaculture.

2000 National Seminar and Exhibition on "Sustainable fisheries and aquaculture for nutritional security" Chennai

56. Medicinal plant leaf extracts as immunostimulants for health management in finfish aquaculture.
57. Immunomodulation by leaf extract of *Acalypha indica* Linn. in *Oreochromis mossambicus* (Peters).
58. Immunostimulation by *Ocimum sanctum* (Linn) leaf extract administered through different routes in *Oreochromis mossambicus* (Peters).
59. Comparative efficacy of crude extract and water soluble fraction of *Murraya koenigii* as immunostimulants in *Oreochromis mossambicus*.
60. Effect of crowding stress and sex ratio on host resistance in *Oreochromis mossambicus*.
61. Differential immunomodulation by chromium and selenium on the antibody response and the neutrophil activity of *Oreochromis mossambicus* (Peters).
62. Studies on the spleen and head kidney of an Indian major carp, *Labeo rohita*.
63. Comparative study of the immunostimulation by crude extract and water soluble fraction of *Piper betle* leaves in *Oreochromis mossambicus*.
64. Studies on temporal relationship between immunostimulant administration and vaccination in *Oreochromis mossambicus*.

1996 National Workshop on Fish and Shellfish Health Management, Central Institute of Freshwater Aquaculture, Bhubaneswar, Orissa

65. Immunostimulants- an alternative to vaccines in finfish health management.
66. Immunostimulatory effect of vitamin E in *Oreochromis mossambicus* (Peters)
67. Immunostimulation by leaf extracts a few South Indian medicinal plants in *Oreochromis mossambicus* (Peters)
68. Immunomodulatory effect of Selenium and Chromium on the antibody response in *Oreochromis mossambicus* (Peters)
69. Immunostimulation by leaf extract of *Ocimum sanctum* in *Oreochromis mossambicus* (Peters)
70. Azadirachtin- an immunostimulant for *Oreochromis mossambicus* (Peters)

1996. 4th Indian Fisheries Forum, Kochi, Kerala

71. Immunological indicators of environmental stress – a field study in *Oreochromis mossambicus* (Peters)
72. Immunosuppressive effect of organic and inorganic forms of Nickel in *Oreochromis mossambicus* (Peters)
73. Immunosuppressive effect of the organic chloride pesticide, Endosulfan in *Oreochromis mossambicus* (Peters)
74. Immunomodulatory effect of Aluminum on the humoral immune response to bovine serum albumin in *Oreochromis mossambicus* (Peters)
75. Effect of the organophosphate pesticide Quinolphos (Ekalux) on the immune response in *Oreochromis mossambicus* (Peters)
76. Immunosuppression by the textile dye, Direct Blue 2B in *Oreochromis mossambicus* (Peters)
77. Immunomodulatory effect of vitamin A in the fish, *Oreochromis mossambicus* (Peters)

78. Immunostimulatory potential of levamisole with reference to antibody response in *Oreochromis mossambicus* (Peters)
79. Immunostimulatory effect of Azadirachtin in *Oreochromis mossambicus* (Peters)

1995 National Seminar on Aquatic Animal Health, University of Kerala, Trivandrum

80. Differential immunomodulatory effect of two different forms of Mercury in *Oreochromis mossambicus* (Peters)
81. Effect of Fenvalerate (Sumicidin) on the humoral immune response in *Oreochromis mossambicus* (Peters)
82. Stress- induced immunomodulation in *Oreochromis mossambicus* (Peters)
83. Effect of Chromium compounds on the cell-mediated immunity in *Oreochromis mossambicus* (Peters)
84. Immunomodulatory impact of Zinc in *Oreochromis mossambicus* (Peters)
85. The effect of Monocrotophos (Nuvacron) on the antibody response to bovine serum albumin in *Oreochromis mossambicus* (Peters)
86. Relative efficacy of immersion and hyperosmotic infiltration methods of immunization in fish

1995 National Research Seminar on Metal Toxicity, K.R. G. College, Gwalior, Uttar Pradesh

87. Modulation of humoral immune response by tannery effluent in *Oreochromis mossambicus* (Peters)
88. Effect of Aluminum Sulfate on the humoral immune response to bovine serum albumin in *Oreochromis mossambicus* (Peters)

1993 The UGC National Seminar on Fish Health, Bharathiar University, Coimbatore, Tamilnadu

89. Differential Effect of Chromium on the antibody response in *Oreochromis mossambicus* (Peters)
90. Effect of Nickel on the humoral immune response to a protein and a cellular antigen in *Oreochromis mossambicus* (Peters)

1990 2nd Indian Fisheries Forum, Mangalore, Karnataka

91. Effect of the steroid hormone, hydrocortisone on the antibody response in the teleost fish, *Oreochromis mossambicus* (Peters)
92. Effect of Carageenan on the antibody response to a protein and a cellular antigen in the fish *Oreochromis mossambicus* (Peters)
93. Role of helper cells in the antibody response to bovine serum albumin in the fish, *Oreochromis mossambicus* (Peters)
94. Studies on the circadian variations in antibody production in the fish, *Oreochromis mossambicus* (Peters)
95. Influence of dietary protein level on the antibody production in the fish, *Oreochromis mossambicus* (Peters)
96. Effect of the heavy metal, cadmium on the antibody production in the fish, *Oreochromis mossambicus* (Peters)

1988 National Seminar on Aquatic Environment and Organisms Chikkiah Naicker college, Erode, Tamil Nādu

97. Effect of the heavy metal, Chromium on the antibody response to a protein antigen in the fish, *Oreochromis mossambicus* (Peters)

98. Effect of Carageenan (Lambda) on the antibody response to a protein antigen in the fish, *Oreochromis mossambicus* (Peters)
99. Effect of Lambda Carageenan on the humoral immune response to sheep erythrocytes in the fish, *Oreochromis mossambicus* (Peters)
100. Immunotoxic effect of the heavy metal, Mercury on the Cichlid fish, *Oreochromis mossambicus* (Peters)

Annexure-6

Invited lectures on Fish Immunology and related areas delivered at international institutions including the following

1. **2001** - Hannover Veterinary University, Hannover, Germany.
2. **1999** - Hannover Veterinary University, Hannover, Germany.
3. **1999** - Oberlin College, Oberlin, Ohio, United States.
4. **1998** - Oberlin College, Oberlin, Ohio, United States.
5. **1996** - Brisbane, Queensland, Australia.
6. **1995** - Hong Kong Baptist University, Hong Kong.
7. **1990** - Bellevue University, Bellevue, Nebraska, United States
8. **2009** - Virginia Institute of Marine Science, Virginia, US
9. **2009** - National Centre for cool and cold water Aquaculture, West **Virginia**, US

Invited lectures on Fish immunology and related areas were delivered in many National institutions including the following

10. Delhi University, New Delhi.
11. Central Institute of Fisheries Education, Versova, Bombay.
12. Central Institute of Fresh water Aquaculture, Bhuvaneswar, Orissa.
13. Goa University, Goa.
14. Kerala University, Kerala.
15. Sri Avinasilingam University, Coimbatore.
16. Bharathidasan University, Trichirappali.
17. Madurai Kamaraj University, Madurai.
18. Karunya University, Coimbatore.

Large number of Post graduate Colleges in Tamil Nadu and neighbouring states.

Annexure-7**Major Research Projects completed in the last 15 years**

No.	Project title	Funding Agency	Total Cost	Duration	Place of Work
1.	Immuno-prophylactic and therapeutic potentials of the hemi-parasitic mistletoe, <i>Dendrophthoe falcata</i> in Asian seabass, <i>Lates calcarifer</i>	Dept of Biotechnology, India	42 Lakhs	2013-17	Vels Institute of Science, Technology and Advanced Studies (VISTAS)
2.	Immunostimulatory, antibacterial and disease protective active principles from selected marine macroalgae for application to aquaculture	Ocean Atmospheric Science and Technology Cell (MoES)	13.2 Lakhs	2008-11	Lady Doak College
3.	Studies on the Immune responses of the Striped murrel, <i>Channa striatus</i>	Dept of Science and Technology	23 Lakhs	2008-11	Lady Doak College
4.	Development of Immunostimulant(s) from <i>Nyctanthes arbor-tristis</i> for application in Aquaculture	University Grants Commission	10.3 Lakhs	2005-08	Lady Doak College
5.	Isolation of Immunostimulatory and antibacterial active principles from selected plants for application medicinal in finfish aquaculture	Dept of Biotechnology, India	20 Lakhs	2003-06	Lady Doak College
6.	Immunological indicators of stress/pollution and of disease outbreak in Indian finfish aquaculture	Volkswagen Foundation, Germany	30 Lakhs	2001-04	Lady Doak College
7.	Development of Fish Vaccines against <i>Aeromonas hydrophila</i> infection	Dept of Biotechnology, India	10 Lakhs	1994-98	The American College
8.	Studies on immunity in the tilapia, <i>Oreochromis mossambicus</i> Peters	United Board, New York, USA	3 Lakhs	1990-92	The American College

Annexure-8**Titles of Ph.D. Theses in Fish Immunology Supervised by Dr. R. Dinakaran Michael**

NO	Doctoral Scholar	YEAR	TITLE
1.	S. Kalaivani Priyadarshini	2021	Studies on the immune response of striped murrel, <i>Channa striatus</i> Bloch
2.	K.S. Shalini	2020	Studies on Immunoprophylactic potentials of Hemiparasitic mistletoe, <i>Dendrophthoe falcata</i> (L.f) Ettingsh on <i>Oreochromis niloticus</i> (Linnaeus)
3.	Omita Yengkhom	2019	Immunostimulatory and disease protective properties of selected-macroalgal polysaccharide fraction in <i>Oreochromis niloticus</i>
4.	A.S. Parasuraman	2018	Immunostimulating and adjuvanting properties of the Indian mistletoe, <i>Dendrophthoe falcata</i> and an opportunistic pathogen, <i>Aeromonas hydrophila</i>
5.	B. Ramalakshmi	2012	Immunostimulatory, antibacterial and disease protective active principles from the selected marine macroalgae for application to aquaculture (currently doing doctoral work)
6.	Priyatharsini Rajasekar	2012	Studies on the immunostimulatory properties of selected Marine macroalga in the major carp, <i>Labeo rohita</i> (currently doing doctoral work)
7.	John Wesly Kirubakaran	2009	Immunomodulatory effect of <i>Nyctanthus arbor-tristis</i> L seed extracts in <i>Oreochromis mossambicus</i>
8.	M. Divya Gnaneswari	2007	Effect of <i>Solanum trilobatum</i> leaves on the immune response of <i>Oreochromis mossambicus</i>
9.	Catherine P. Alexander	2007	Studies on the immunomodulatory response of <i>Tinospora cordifolia</i> Merris leaves in <i>Oreochromis mossambicus</i> .
10.	L.D. Devasree	2006	Immunomodulatory effect of extracts of <i>Nyctanthes arbor-tritis</i> L. leaves in <i>Oreochromis mossambicus</i> (Peters)
11.	C. Binu Ramesh	2005	Confinement stress-induced immunomodulation in <i>Oreochromis mossambicus</i>
12.	M. Prabakaran	2005	Evaluation of immunomodulatory effect of chromium in <i>Oreochromis mossambicus</i>

Annexure-9**Titles of M. Phil. dissertations in Fish Immunology Supervised by Dr. R. Dinakaran Michael**

NO	M.Phil. Scholar	YEAR	TITLE
1.	M. Shabana Parveen	2008	Immunostimulatory, Antibacterial and Disease Protective for properties of the selected bioactive fractions (F1, F6, F7 & Cp), of the Phaeophycean Macroalga, CF1-MA-01 in <i>Oreochromis mossambicus</i> (Peters)
2.	T. Pabitha	2008	Effect of the selected Bioactive Fractions (F4, F5 & F7) of the Phaeophycean Macroalga, CF1-MA-02 in <i>Oreochromis mossambicus</i> (Peters)
3.	S. Kalaivani Priyadarshini	2007	Immunostimulatory, Antibacterial and Disease protective properties of a phaeophycean macroalga (CFI-MA 01) extracts in <i>Oreochromis mossambicus</i> (Peters)
4.	A. Sujatha	2007	Immunomodulatory and Antibacterial activity of <i>Dendrophthoe falcata</i> (Ettingsh) extracts in <i>Oreochromis mossambicus</i> (Peters)
5.	P. Arumugam	2004	The effect of <i>Andrographis paniculata</i> (Nees) leaf extracts on specific and non specific immune response in <i>Oreochromis mossambicus</i> (Peters)
6.	S.P. Balasubramani	2002	Immunomodulation by the fruit extract of Indian gooseberry, <i>Phyllanthus embica</i> (LINN) in <i>Oreochromis mossambicus</i> (Peters)
7.	F. Anitha	2002	Effect of crowding as a social stress of immune response and serum cortisol level in <i>Oreochromis mossambicus</i> (Peters)
8.	V. Brinda devi	2001	Comparative efficacy of crude extract and water-soluble fraction of <i>Murraya koenigii</i> as immunostimulants for <i>Oreochromis mossambicus</i> (Peters)
9.	A. Vanitha	2001	Immunostimulatory effect of crude extract and water-soluble fraction of <i>Piper betle</i> leaves in <i>Oreochromis mossambicus</i> (Peters)
10.	P. Subbulakshmi	2000	Studies on the head kidney, spleen and peripheral blood cells of the Indian Major carp <i>Labeo rohita</i>

11.	P. Venkatesan	2000	Immunomodulation by leaf extracts of <i>Ocimum sanctum</i> Linn. and <i>Crossandra infundibuliformis</i> administered by immersion in <i>Oreochromis mossambicus</i> (Peters)
12.	Albert Arul Daniels G	1998	Vaccination with whole cell bacterial antigen of <i>Aeromonas hydrophila</i> in <i>Oreochromis mossambicus</i> (Peters)
13.	YC. Viji	1998	Protection of <i>Oreochromis mossambicus</i> (Peters) by leaf extract of <i>Ocimum sanctum</i> Linn. against <i>Aeromonas hydrophila</i> experimental infection.
14.	S. Venkatalakshmi	1996	Immunomodulation by leaf extract of <i>Ocimum sanctum</i> Linn. in <i>Oreochromis mossambicus</i> (Peters)
15.	S.M. Logambal	1996	Immunostimulatory effect of Azadirachtin in <i>Oreochromis mossambicus</i> (Peters)
16.	M. Muthulakshmi	1996	Immunostimulatory effect of the leaf extract of <i>Phyllanthus niruri</i> on the immune response in <i>Oreochromis mossambicus</i> (Peters)
17.	K. Radhakrishnan	1995	Immunosuppressive effect of organic and inorganic forms of Nickel in <i>Oreochromis mossambicus</i> (Peters)
18.	S. Anitha Caroline	1994	Relative efficacy of different methods of immunization
19.	Richard B. Rajan	1993	Immunostimulation by vitamin A in <i>Oreochromis mossambicus</i> (Peters)
20.	M. Rajesh	1993	Immunostimulatory effect of Aluminum in <i>Oreochromis mossambicus</i> (Peters)
21.	T. Sudhan	1993	Immunostimulatory effect of vitamin E in <i>Oreochromis mossambicus</i> (Peters)
22.	P. Beulah Rajavarthini	1993	Isolation and Characterization of immunoglobulins in <i>Oreochromis mossambicus</i> (Peters)
23.	Lewis Marian Agnella	1992	Modulatory effect of the organic and inorganic forms of Mercury on the immune system of <i>Oreochromis mossambicus</i> (Peters)
24.	Priyadharsini Rajasekaran	1992	Differential effect of trivalent and hexavalent forms of Chromium on the immune system of <i>Oreochromis mossambicus</i> (Peters)

25.	Radhika Rajan	1990	Effect of the heavy metal Zinc on the antibody response to protein and cellular antigens in <i>Oreochromis mossambicus</i> (Peters)
26.	R. Geetha	1990	Studies on the effect of the heavy metal Lead on the humoral immune response to the protein antigen, bovine serum albumin in <i>Oreochromis mossambicus</i> (Peters)
27.	Gracelet D. Thampidhas	1989	Effect of the heavy metal, Mercury on the antibody response and oxygen consumption in the teleost fish, <i>Oreochromis mossambicus</i> (Peters)
28.	V. Navarathina Jothi	1989	Effect of the heavy metal Lead on some physiological aspects of the Cichlid fish, <i>Oreochromis mossambicus</i> (Peters)
29.	S. Muralitharan	1989	Effect of Kappa, Lambda and Iota Carageenan on the antibody response in <i>Oreochromis mossambicus</i> (Peters)
30.	P. Victoria Thangam	1989	Modulation of antibody response by the heavy metal Mercury in <i>Oreochromis mossambicus</i> (Peters)
31.	C. R. Manonmani	1988	Effect of Carageenan on the antibody response in the fish, <i>Oreochromis mossambicus</i> (Peters)
32.	A. Meenal	1988	Effect of the heavy metal Cadmium on the antibody response to a protein and a cellular antigen in the fish, <i>Oreochromis mossambicus</i> (Peters)

Annexure 10

Membership in Science Professional Societies

- World Aquaculture Society, Baton Rouge, LA, USA
- Fish Health Section (WAS), USA
- American Fisheries Society, Bethesda, MD, USA
- International Society for Fish and Shellfish Immunology
- Asian Fisheries Society, Kuala Lumpur, Malaysia
- Fish Health Section (AFS), Malaysia
- Asian Fisheries Society, Indian Branch, Mangalore, India